🧠 AI Evaluation Methodology & Protocol

1. Overview

This document outlines the standard operating procedures (SOPs) I use for auditing, benchmarking, and evaluating Large Language Models (LLMs). My approach is adapted from rigorous **Root Cause Analysis (RCA)** and **Six Sigma** methodologies traditionally used in mechanical engineering.

2. Core Principles

- **Zero AI-Assistance Policy:** All evaluations, rationales, and critiques are generated 100% by human reasoning. Using AI to evaluate AI introduces recursive bias and degrades the dataset quality.
- **Empirical Validation:** Output is not just judged on "fluency" but on logical consistency, factual accuracy, and domain-specific rigor.
- **Implicit Gap Analysis:** Identifying where an LLM makes hidden assumptions or skips logical steps, akin to finding hidden sequence logic failures in mechanical assemblies.

3. Evaluation Framework

1. **Fact-Checking & Sourcing:** Cross-referencing technical claims against established academic literature and engineering principles.
2. **Reasoning Traceability:** Can the model's logic be traced step-by-step? (Similar to tracing panel datum points in physical manufacturing).
3. **Edge-Case Stress Testing:** Pushing the model to failure points to understand its boundaries and failure modes.

4. Rationale Structuring

When writing feedback rationales for AI labs, I employ a structured format:

- **The Error:** Precise identification of the flaw (e.g., hallucination, logical leap).
- **The Root Cause:** *Why* the model likely failed (e.g., misinterpreting the prompt's constraints).
- **The Correction:** The empirically correct answer with citations or step-by-step logic.